

Caries segmentation on tooth X-ray images with a deep network

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ARTICLE INFO

Keywords:

Dental caries
Segmentation
Deep learning
U-shaped network
X-ray
Medical image
Artificial intelligence

ABSTRACT

Objectives: Deep learning has been a promising technology in many biomedical applications. In this study, a deep network was proposed aiming for caries segmentation on the clinically collected tooth X-ray images.

Methods: The proposed network inherited the skip connection characteristic from the widely used U-shaped network, and creatively adopted vision Transformer, dilated convolution, and feature pyramid fusion methods to enhance the multi-scale and global feature extraction capability. It was then trained on the clinically self-collected and augmented tooth X-ray image dataset, and the dice similarity and pixel classification precision were calculated for the network's performance evaluation.

Results: Experimental results revealed an average dice similarity of 0.7487 and an average pixel classification precision of 0.7443 on the test dataset, which outperformed the compared networks such as UNet, Trans-UNet, and Swin-UNet, demonstrating the remarkable improvement of the proposed network.

Conclusions: This study contributed to the automatic caries segmentation by using a deep network, and highlighted the potential clinical utility value.

1. Introduction

Oral diseases have been estimated to affect nearly half of the global population nowadays, among which 2.3 billion people are suffering from permanent dental caries worldwide [1]. The clinical treatment for caries is still limited in terms of the fact that once a severe degree of caries has developed it is impossible to be remineralized and only destructive extraction can be performed. Therefore, it is particularly important to detect dental caries in the early stage and prevent the further decay process. However, an effective diagnosis of dental caries is still difficult, and thus much focus is gained in international oral researches.

Traditional clinical detection of dental caries mainly relies on dentists' visual examination [2], which depends on an individual's experience and is highly subjective, and as a result prone to erroneous judgment. Moreover, early caries are hard to be distinguished with naked eyes, especially those occurring in the areas such as the adjacent surface of teeth. X-ray imaging is thought to be the standard for detecting caries that is invisible to the naked eye, and it also helps in understanding the depth of caries. However, a stomatologist's

experience in reading X-ray images will affect the caries detection result, as young dentists with less experience might lead to missing detection of some non-obvious caries. On the other hand, examining abundant X-ray images also increases the work of doctors. Therefore, an automatic detection method with high reproducibility and accuracy is highly demanded to provide objective caries diagnostic assistance for clinical stomatologists.

Some researchers have focused on the computer-aided detection of caries with X-ray images [3–6]. Tracy et al [3]. evaluated and confirmed the effectiveness of a density analysis aiding tool in improving dentists' ability to detect and classify caries. Oliveira and Proença [4] presented the detection of dental caries in panoramic dental X-ray images by using selected features, and in their method, the feed-forward artificial neural network was used for classifying if caries exist. Recently Osterloh and Viriri [5] also proposed an unsupervised learning model for caries detection and achieved a good performance. However, all of these studies conducted only detection of caries and no segmentation was concerned. Caries segmentation is also important because it provides more information about the decay degree, such as the decayed area, which could be the precondition for further caries classification. Besides,

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most of the previous studies adopted traditional image process methods such as thresholding to detect caries, which may be unstable when the image attribute (brightness, etc.) changes.

With rapid development, deep learning has been widely used in various medical imaging tasks and achieved remarkable success [7,8]. Compared to traditional methods, deep learning has the advantage of being more robust to complex interference owing to its powerful self-extraction capability of feature representation and flexibility of network architecture. A few researchers tried to segment teeth by using deep convolutional neural networks (CNNs) [9–11], and promising results were confirmed. Recently Lee et al. [12] also evaluated the efficacy of the GoogLeNet deep CNN network for dental caries detection on periapical radiographs. However, there are still very few researches reporting the deep-learning-based caries segmentation till now. Wang et al. [13] claimed a poor precision for caries segmentation by Ronneberger et al.'s deep CNN method [11]. Recently, Cantu et al. [14] used UNet to detect caries lesions on bitewings and found it significantly more accurate than dentists. Lee et al. [15] also adopted UNet for early caries detection in bitewing radiographs and concluded that the deep model could help clinicians to diagnose dental caries more accurately. Although the segmentation UNet was used in these studies, the researchers only focused on caries-level detection, and no segmentation metrics were reported. Along with the rapid rise of artificial intelligence researches in dentistry, Schwendicke et al [16], lately released a checklist for authors, reviewers, and readers to facilitate the standards in this field. Caries segmentation is still a hard task even for deep learning methods. The tooth X-ray image is usually interferential and with low contrast, making it difficult to segment the decayed area from the background. In addition, caries lesions are of diversified scales, and early lesions are very subtle to be recognized, which is another difficulty for caries segmentation.

Considering the lack of studies of caries segmentation and its difficulty, we proposed a new deep network to segment caries on tooth X-ray images in this paper. We inherited the encoder-decoder architecture and skip connection idea used in the U-shaped deep network [11]. To overcome the inherent locality limitation of CNN, a transformer embedding module with a global attention mechanism presented by Chen et al. [17] was also adopted in our network. Furthermore, we introduced the atrous spatial pyramid pooling (ASPP) to enlarge the receptive field to enhance the network's global feature extraction capability. In addition, the feature pyramid method [18] was employed during decoding to realize the cross-layer feature fusion, which could effectively deal with the multi-scale issue of caries. We compared our method with classical networks such as UNet [11] and UNet++ [19]; on the same dataset. Yielded results demonstrated the outperformance of our network in segmentation and good robustness.

2. Method

2.1. Tooth X-ray image collection in clinic

The tooth X-ray image collection was approved by the bioethics committee of our hospital (No. 2021-58R). The X-rays in the study were taken from the patients visiting the prosthodontics department who needed an X-ray of the root tip for dental caries or other reasons during the year 2020–2021. The ethnicity of all patients is yellow race. All of the patients signed an informed consent form. The inclusion criteria were: teeth with obvious or suspected caries on the adjacent surface, no developmental abnormalities, no prostheses, no malocclusion, and no periodontal diseases. While the exclusion criteria were: patients with severe periodontal diseases, irregular dentition, or dentition loss; patients with mental diseases, critical diseases, pregnant women, and other uncooperative patients. This study does not involve vulnerable groups. There were 53% male and 47% female patients in the dataset. The ages were in the interval from 20 to 79 years. All of the data were generated using the radiographic machine FOCUS from the manufacturer

Instrumentarium Dental (Tuusula, Finland). Finally, 153 radiographs of the root tip were clinically collected as the initial dataset.

2.2. Sample size estimation

To explore the sample size of the test dataset, we adopted the sample size estimation formula for the single-arm diagnostic test with objective performance criteria [20].

$$n = \left(\frac{Z_{1-\alpha/2} \sqrt{p_0(1-p_0)} + Z_{1-\beta} \sqrt{p(1-p)}}{p - p_0} \right)^2 \quad (1)$$

Where $Z_{1-\alpha/2} = 1.96$ and $Z_{1-\beta} = 1.28$ could be determined by the look-up table with a significance level of $\alpha = 0.05$ and the study power of $1-\beta = 0.9$. The expected sensitivity p and the minimum standard of clinically acceptable sensitivity p_0 were assumed to be 0.80 and 0.60, respectively. A sample size of 55 teeth was then obtained according to the formula. Therefore, considering each radiograph containing about 3 teeth, at least 19 X-ray radiographs should be included in the test dataset. To have enough margin, the test dataset was determined to be doubled the calculation result, which was finally 40 radiographs.

2.3. Labeling and data augmentation

All of the collected tooth X-ray images were independently labeled by two professional stomatologists with more than 10-year clinical experience, with the caries decay area delineated meticulously. A senior stomatologist with more than 20-year clinical experience was then in charge of reviewing and revising all the labels. The software LabelMe [21] was adopted to annotate the images. All of the experts were trained on how to use the annotation tool before the labeling task. Afterward, 40 images were randomly selected out of the whole as the test data set that is independent of the training data set. Due to the limitation of available training images, the remaining 113 images were then augmented to form a larger training data set.

The applied data augmentation is the popular and conventional offline method in deep learning, including translation, flip, mirror, rotation, vertical perspective deformation, oblique quadrangular perspective deformation, region warping, and shear mapping transformation for the images and their labeled masks. We do not use online dynamic data expansion mainly due to its uncontrolled randomness of the training process, which is highly likely to produce meaningless enhanced images thus resulting in wasted training resources and useless generalization parameters. These methods applied in this study are effective in enhancing sample diversity regarding imaging angles, tooth shape, tooth size, and caries area, which leads to better generalization performance of the trained model. Concretely, the implementation of the enhancement is to apply the above-mentioned transform operations with the help of a library package named Augmentor [22], where the rotation angle is set between -15° and 15° and the perspective deformation parameter magnitude is a random number in the interval of (0, 1). All the augmented images were then validated by the labeling dentists. A total number of 800 images were generated and used in the network training process finally, of which 10% of images were used as validation during training sessions.

2.4. Deep network architecture

Our proposed deep network for pixel-level caries segmentation draws on the implementation of UNet [11] and Trans-UNet [17]. The complete network architecture is depicted in Fig. 1. In the encoder part, a ResNet variant is adopted as the backbone for feature extracting. A visual transformer module is embedded after the backbone to consolidate the linkage of contextual information, while an Identity-ASPP (In-ASPP) module ends the encoding process to further capture the multi-scale global information. The In-ASPP modules are also used in the

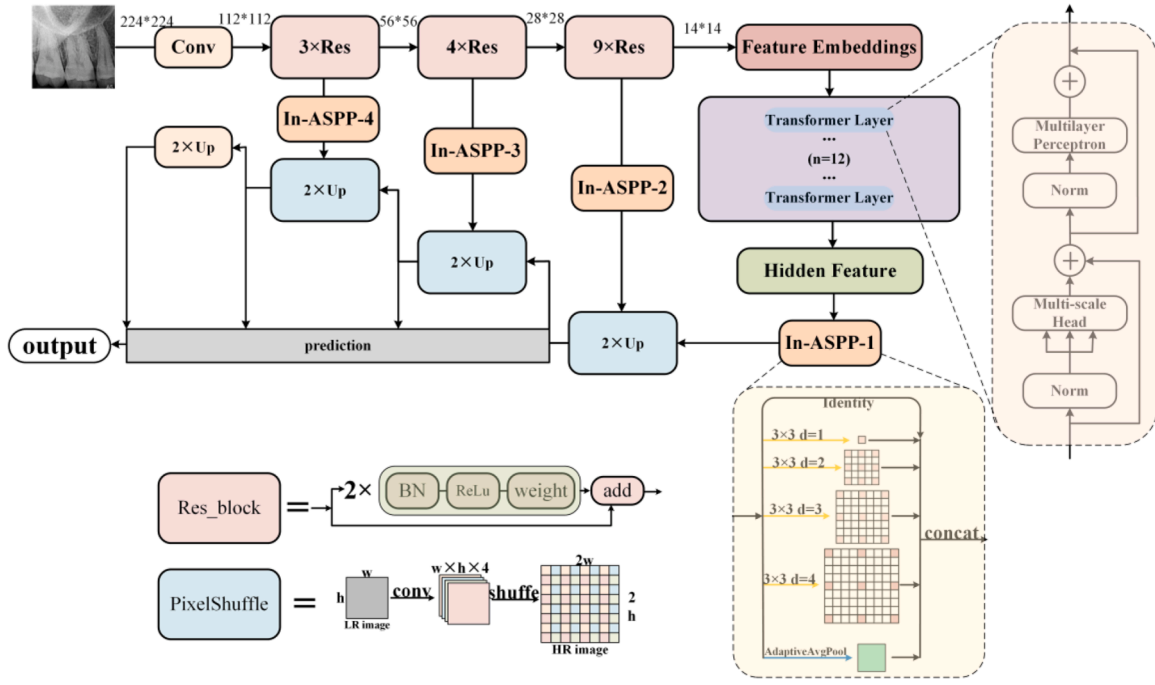


Fig. 1. The proposed deep network architecture for caries segmentation.

skip connections between the encoding and decoding, as shown in the figure. This configuration is in an attempt to sample the different resolution feature layers in parallel with the dilated convolution at different sampling rates. In the decoding part of the network, we use the more efficient “PixelShuffle” method for upsampling. Instead of only using the last decoding layer for prediction, we combine the various decoding layers to form the feature pyramid network (FPN) for better feature fusion, and at last resulting in a segmentation output.

For backbone, we use the 50-layer ResNetv2 as the feature extractor for the following encoding process, which is based on an improvement of the initial ResNet [23]. In each residual block, the batch normalization (BN) layer is used as pre-activation, which not only plays the role of regularization but also avoids the non-negative influence of the ReLU activation function on feature expression in the forward process. As shown in Fig. 1, the residual block extracts four copies of the features, in which the first three are used for skip-connections, and the remaining depth features are used to establish long-range dependencies on the image information.

The encoding ends with the Transformer module and the In-ASPP module, both of which are used to obtain broader contextual information in a broad sense. The Transformer for sequence-to-sequence prediction processes the convolutional feature map after the image block has been encoded and labeled in a global attention manner, i.e., forming a linear embedding sequence in the form of a patch as input and a patch of the same sequence length as output. After that, the features extracted from multiple-scale receptive fields through the ASPP [24] are then connected in parallel with the adaptive pooling of features and itself, as shown in Fig. 1, forming the so-called In-ASPP module. Four different dilation rates in each In-ASPP module ensure the fusion of multi-scale information. The similar In-ASPP modules are used in the skip connections between encoder and decoder, as tagged as In-ASPP-k in Fig. 1, where k means a corresponding multiplier of the dilation rates in each In-ASPP module to meet the needs of the receptive field at corresponding feature map sizes.

The main function of the decoding process is the upsampling of features. Rather than the usual interpolation, we take the PixelShuffle method that was originally used in image super-resolution reconstruction [25] for upsampling, which will be more efficient. Concretely, the

low-resolution (LR) $w \times h$ image is convolved in a series to obtain an r^2 number of feature maps, where r is the multiple of the upsampling, and then periodic shuffled to obtain the final high-resolution (HR) image with a size of $rw \times rh$. Each upsampled feature map is concatenated with the feature map from the skip connection path to form a decoder feature layer, which is finally fused by the FPN method [26] to yield an accurate pixel classification result.

In summary, taking the input X-ray image of size $H \times W \times C$ as an example, the execution process of the network can be summarized as follows. In the encoding flow, the ResNet-v2 backbone extracts multi-level features with the help of residual modules and different strides, which is then followed by the image serialization prepared for the Transformer module. The feature map is sliced to $N = (H \times W)/P^2$ patches of $P \times P$ size in image serialization, which is then mapped to patch embeddings by a linear projection [27]. By adding position embeddings, the resulting sequence of embedding vectors serves as input to the transformer layer. The output from the Transformer module is fed into the In-ASPP-1 module for further pooling operations. After that, it is entered into the decoding process. There are four upsampling with the PixelShuffle method to generate the same size output as the input image. The first three upsampling are followed by the fusion with the skip connection features that have been passed through the In-ASPP module from the corresponding encoder layer. At last, the constructed FPN that utilizes the features of all decoder layers will obtain the pixel segmentation result of caries.

2.5. Evaluation metrics

We use the dice similarity and precision to evaluate the networks' performance. Considering the image segmentation as the pixel-level classification problem, TP (True Positives), TN (True Negatives), FP (False Positives), and FN (False Negatives) are defined for every pixel according to its classification results. The pixel classification precision, i.e., the accuracy rate at which the model is true in the prediction, is expressed as,

$$precision = \frac{TP}{TP + FP} \quad (2)$$

The dice similarity quantifies how closely the network's segmentation result matched the hand-annotated ground truth mask. If A is defined as the set of manually masked pixels, and B is the set of network segmented pixels, then the dice similarity can be defined as the following formula.

$$dice = \frac{2|A \cap B|}{|A| + |B|} = \frac{2TP}{2TP + FN + FP} \quad (3)$$

Besides the above segmentation metrics, we also provide caries-level detection metrics that are concerned in clinical. According to the tooth level statistics method in the literature [14], caries detection results are defined as tp, fp, tn, and fn (to distinguish from the pixel-level statistics lower cases are used herein) based on overlapping between the model prediction blob and the corresponding ground truth mask. Sensitivity, also known as the true positive rate (tpr) which represents the percentage of cases that are diseased to be correctly diagnosed, and specificity, also known as the true negative rate (tnr) which represents the chances of non-misdiagnosis, are calculated as the following expressions.

$$tnr = \frac{tn}{tn + fp} \quad (4)$$

$$tpr = \frac{tp}{tp + fn}$$

2.6. Experimental implementation setup

The experimental software platform environment of this study was Ubuntu18.04 + CUDA10.1 + cudnn7.4.2, and Pytorch 1.8.1 deep learning framework was used for network construction, with Python as the interface language. In the training process, the input image was resized to 224×224 . The batch size was set as 12, and the initial learning rate was 0.005 with a linear variation of decay. We adopted the sum of equal weights of cross-entropy and dice as the loss function, and chose SGD as the optimization algorithm with the momentum and weight decay parameters as 0.9 and 0.0001, respectively. A total number of 150 epochs was set for training. At the same time, we also conducted experiments for other common U-shaped series networks, including UNet, UNet++, Trans-UNet, and Swin-UNet, for comparison in a consistent experimental environment. It is pointed out that all of the networks were pre-trained with ImageNet.

3. Results

The normalized loss curves of the proposed network and compared U-shaped networks during the training process are depicted in Fig. 2,

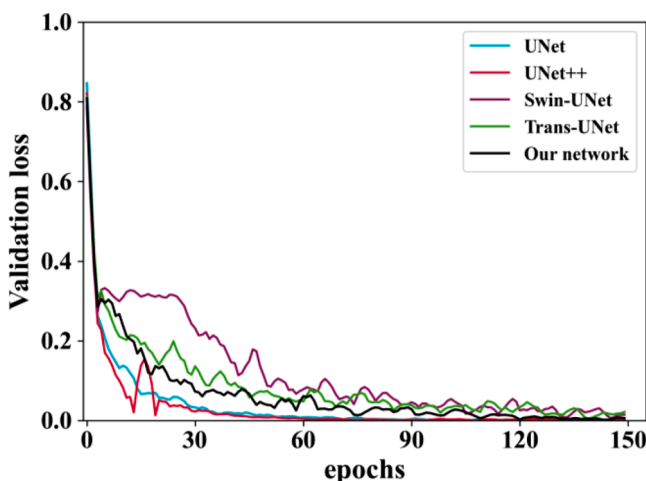


Fig. 2. Normalized test loss diagram of our network and compared networks.

from which we can find that all the networks have reached the convergence state at the end of training. The loss curve of our network converges a little faster than Trans-UNet and Swin-UNet, however slower than UNet and UNet++ which have no Transformer structure inside. The loss of our network finally remains at the same low level as UNet and UNet++, a little bit lower than Trans-UNet and Swin-UNet.

After training, we performed caries segmentation on the test image set using the trained networks. The specific evaluation metrics as given in the above 2.5 section were calculated for each network based on the segmentation results and were listed in Table 1. It can be seen from the table that among all the networks the average dice similarity of our network for the test dataset reaches the highest value of 0.7487, which is higher than not only the Trans-UNet combined with convolution and Transformer but also other U-shaped networks composed of pure convolution (UNet, UNet++) or pure Transformer (Swin-UNet). Referring to the precision metric, our network also achieves the best result, which further demonstrates the superior performance of our network. Compared to the original UNet, a percentage improvement of 19.8 and 27.5% is made for dice similarity and precision by our method.

Fig. 3 shows some example caries segmentation results of our network and typical U-shaped networks on the test tooth X-ray image dataset. The first row shows the tooth X-ray images and the corresponding caries masks, and the rest rows show the segmentation outputs of all the five networks, with one row for one network. It can be noticed that the caries mask is highly variable in shape and area, and the location distribution is not regular. The proposed method is sensitive to boundary capture. Compared with other methods, not only the contour is relatively smooth, but also the segmentation result is the most accurate, while the Trans-UNet segmentation is relatively conservative, and the Swin-UNet is not friendly to caries boundaries. As shown in the figure, segmentation contours of Swin-UNet are relatively rough and have an obvious "sawtooth effect". The rest methods can roughly obtain the caries position, but with poor segmentation effect. However, no algorithm can capture the segmentation boundary perfectly, including our method. Nevertheless, it is worth mentioning that our method can obtain a segmentation precision of more than 0.7, which means having the least number of misclassified pixels. It is emphasized that the precision herein is with pixel classification, not caries detection. Our network could find almost all of the caries if not considering the perfect region overlapping. By counting the detection results of the total 115 teeth in the 40 test images and calculating according to the expression (4), the sensitivity and specificity of the proposed method in dental caries detection are revealed to be 0.94 and 0.92, respectively.

In addition, we plotted the scatter diagram with the dice similarity of each test image for our network and typical U-shaped networks, as shown in Fig. 4. Comparing the fluctuations of dice similarities, it can be seen that the proposed method has the most stable performance in the test image set, with the fewest points lower than 0.5 dice similarity, demonstrating its strong generalization Fig. 5. shows the evaluation metrics of dice similarity and precision on each test image of our method. These metrics show a synergistic trend and are relatively stable without extreme low values, indicating that the proposed method performs well in testing.

4. Discussion

In this study, we tried the segmentation of caries on our own tooth X-rays image dataset collected clinically by using a deep network. The

Table 1

Performance comparison of our network and typical U-shaped networks on the test tooth X-ray image dataset.

	UNet	UNet++	Trans-UNet	Swin-UNet	Our network
<i>dice</i>	0.6251	0.5784	0.7096	0.7076	0.7487
<i>precision</i>	0.5837	0.5441	0.6837	0.7239	0.7443

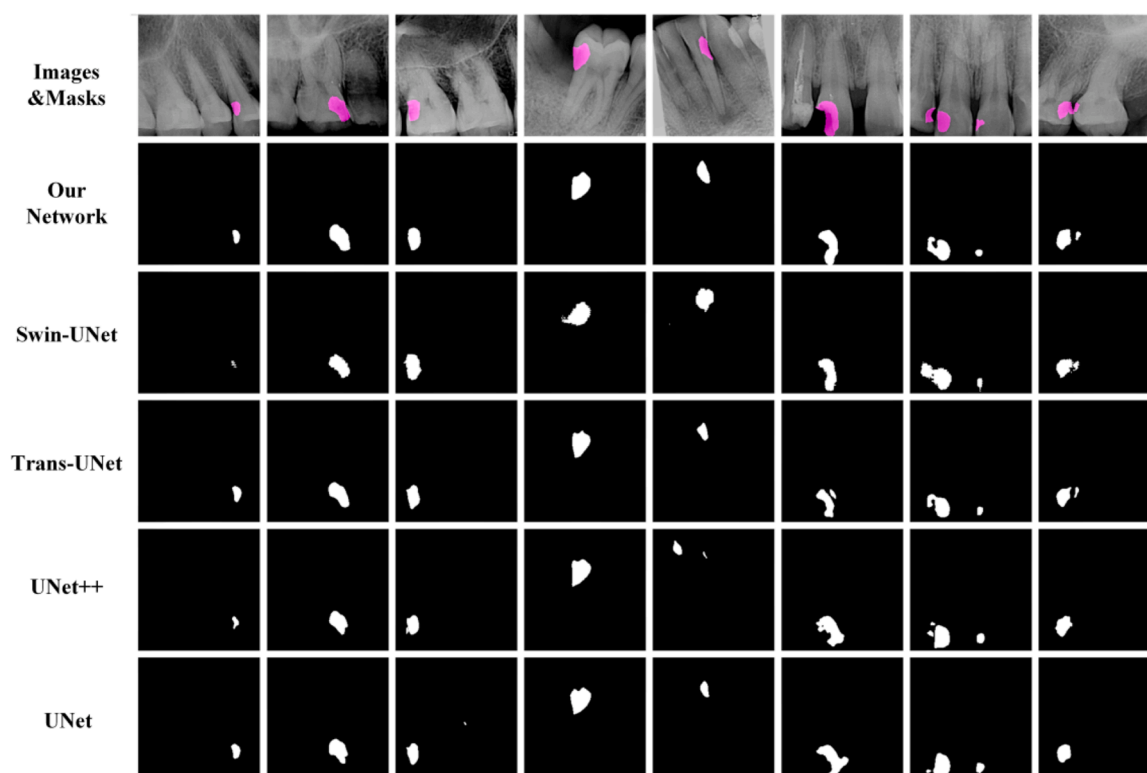


Fig. 3. Example caries segmentation results of our network and typical U-shaped networks on the test tooth X-ray image dataset.

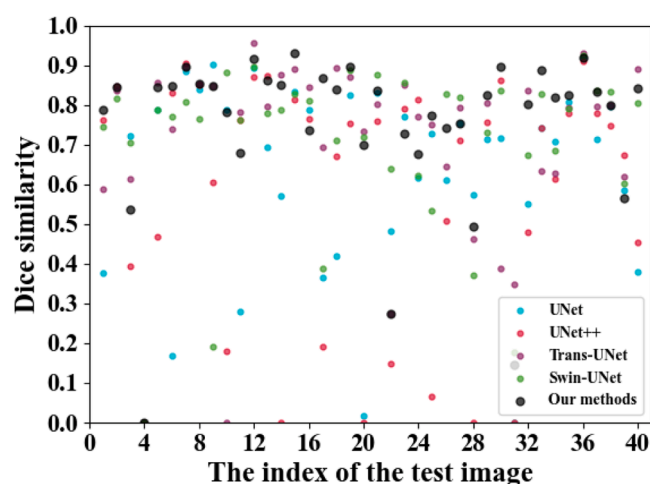


Fig. 4. The scatter plot of each test image's dice similarity for our network and typical U-shaped networks.

proposed network was inherited from the famous U-shaped net series and further improved by using dilated convolution and feature pyramid methods, which were both focused on enhancing the multi-scale and global feature extraction capability together with the vision transformer method. We compared our method with the main previously reported U-shaped series networks, concretely the UNet, UNet++, Trans-UNET, and Swin-UNET, on the same tooth X-rays image dataset. Experimental results demonstrate our network outperforms the other networks in terms of dice similarity and precision metrics on this caries segmentation task.

To our knowledge, although some works report the detection of caries, there are very few researches focusing on caries segmentation. For detection, it is enough to find if caries exist and their rough location, while segmentation needs to outline decay regions that is thought to be a

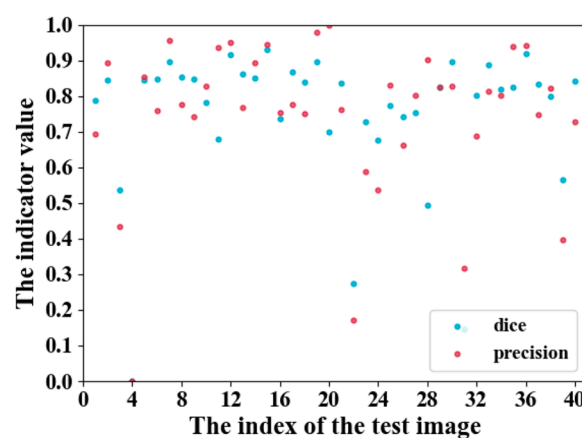


Fig. 5. The dice similarity and precision on each test image of our method.

more difficult task. Ronneberger et al. tried caries segmentation as a part of their distinguished research of UNet, however, the precision and dice similarity were both less than 0.2 [11]. Compared with other tooth structures such as enamel segmentation result that could reach a dice similarity of more than 0.6, this caries segmentation result is quite unsatisfactory, which confirms its difficulty from the side. Since then and till now, there almost no progress published regarding caries segmentation. Our study systemically assesses the possibility of caries segmentation with deep networks for the first time, filling in the gap of recent related researches.

According to our result, the average dice similarity and precision are both around 0.75, which are much higher than those reported in the study of Ronneberger et al. [11]. However, it is noted that the direct metric value comparison is not reasonable because the two studies are with totally different datasets, training and testing sample sizes, data augmentation methods, etc. In our study, a higher dice similarity is

obtained with the same UNet, which is thought to partially benefit from the treble larger training size. However, by comparing the results of UNet and our network in the current study, there is also a 19.8% increase in dice similarity, which demonstrates a huge improvement of our network over the UNet. Compared with the newest Trans-UNet and Swin-UNet, our network still has certain advantages.

Trans-UNet, Swin-UNet, and our network all contain Transformer structures. According to Tay et al. [28], the gain of pre-training for Transformer is greater than that of convolution. To explore how many benefits could get from pre-training for these studied networks, we conducted a comparison between the pre-trained and no pre-trained configurations. According to the evaluation metrics, our network outperforms the other networks whether pre-trained or not. It can be found that the higher the degree of Transformer, such as Swin-UNet, the worse the performance without pre-training, which indicates that the long-range dependence of Transformer is built on a large number of already optimized parameters. To reduce the demand for pre-training, it is a feasible way to combine traditional convolution with Transformer instead of pure Transformer.

In actuality, the dose may affect the features of the X-ray image. To demonstrate the robustness of our method in this situation, we further carried out the segmentation test after manually altering the features of the X-ray images, including brightness variation, image blurring, and pepper & salt noise. It can be found that the segmentation boundary becomes slightly ambiguous after the image features are destroyed, yet the caries regions can still be segmented out without significant effects. The dice similarities and precisions of the test dataset under darkening, brightening, blurring, and pepper & salt noise conditions were found to be 0.7484, 0.7526, 0.7440, 0.6983 and 0.7456, 0.7479, 0.7375, 0.6875, respectively. It can be seen that the pepper & salt noise had the greatest influence. Compared to the original results, the mean dice similarity of all the robustness tests decreased by only 1.29% while the mean precision decreased by only 1.47%. The inference time in the robustness experiments was also tested, and a mean time of 0.059 s/image was found, which increased very slightly compared to the original time 0.058 s/image. It demonstrates that our network can resist environmental interference and noise, and has good robustness and generalization performance.

As this was not a clinical comparison study, we used the objective performance criteria method for the single-arm diagnostic test to determine the test dataset size. An expected sensitivity of 0.80 was assumed in the sample size calculation, which was fulfilled well as the resulting sensitivity achieved more than 0.9 in caries detection. On the other hand, we also conducted experiments to confirm if the training dataset in our study was sufficient. Different numbers of the original

caries images (20, 40, 80, 100 and 113 images) were augmented to the same 800 images to train the same network. The network's performance increases along with the number of original images but becomes saturation after about 80 original images. It indicates that the current training dataset size is sufficient to train our deep learning model.

With good results, our study could be used for the automatic segmentation of caries on tooth X-ray films. Caries segmentation not only finds caries on the tooth but also provides more details of this region. It can be used for doctor-patient communication, allowing patients to have a more intuitive understanding of their caries. And it also has applications in clinical teaching to guide young physicians to better identify caries regions in X-ray images. Due to the low resolution of X-ray images, inexperienced dentists may not clearly distinguish the caries region from the surroundings, while automatic segmentation could offer good support for them. For experienced dentists, it also helps in lessening their work. On the other hand, caries region segmentation provides details, such as the caries region's relative position with the tooth structure, which is the basis for the further accurate classification of caries. In addition, as Cantu et al [14] and Lee et al. [15]. found that UNet was helpful in clinical caries detection, our network could be a good supplement in this clinical application due to its outperforming of UNet.

Although having a promising application clinically, there are still some limitations in our study. The dataset number is small due to the clinical collection limitation. Although the data augmentation method is used to enlarge it, it does not provide extra forms of caries. While the sample size estimation indicates our test dataset is sufficient for the current study, more tooth X-ray images are still desired to further validate the performance of our network. Besides, although reaching high metrics, it is still not enough for clinical usage. The network structure needs to be improved along with a larger dataset to promote the clinical application. Thirdly, we will try to further classify the caries types (shallow, deep caries, etc.) along with the accurate tooth structure segmentation.

5. Conclusions

Caries segmentation is useful in the dental clinic and its automation has become a challenging task. We tried to adopt the hot deep learning technology to solve this issue in this study. We collected the tooth X-ray images in our hospital to form a dataset for our research. By systematically reviewing the U-shaped series networks, we made further improvements to put forward our network. A vision Transformer module, an ASPP module using dilated convolution, and a feature pyramid fusion module were together embedded into the base U-shaped network, in an attempt

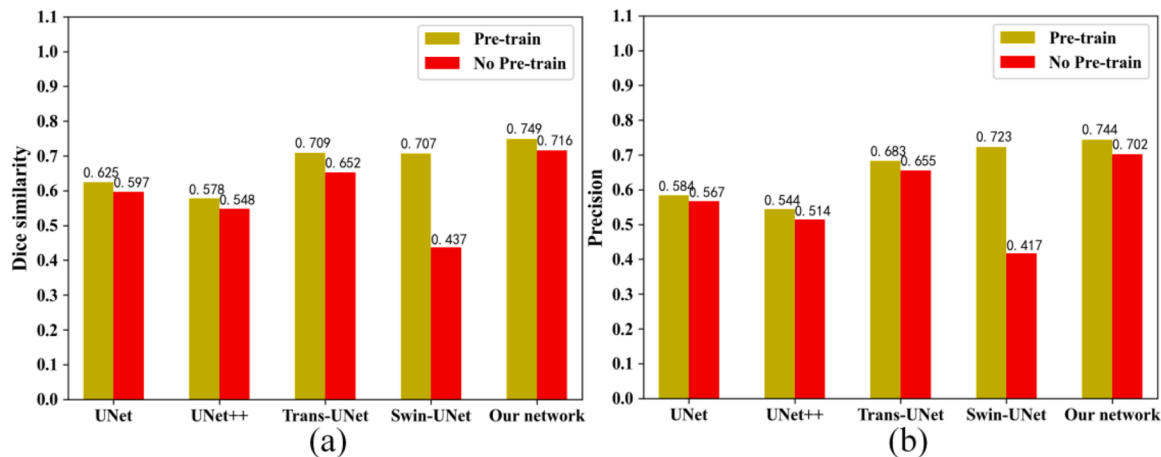


Fig. 6. The effects on segmentation metrics of pre-trained and no pre-trained configurations for UNet, UNet++, Trans-UNet, Swin-UNet, and our network. (a) Dice similarity; (b) Precision.

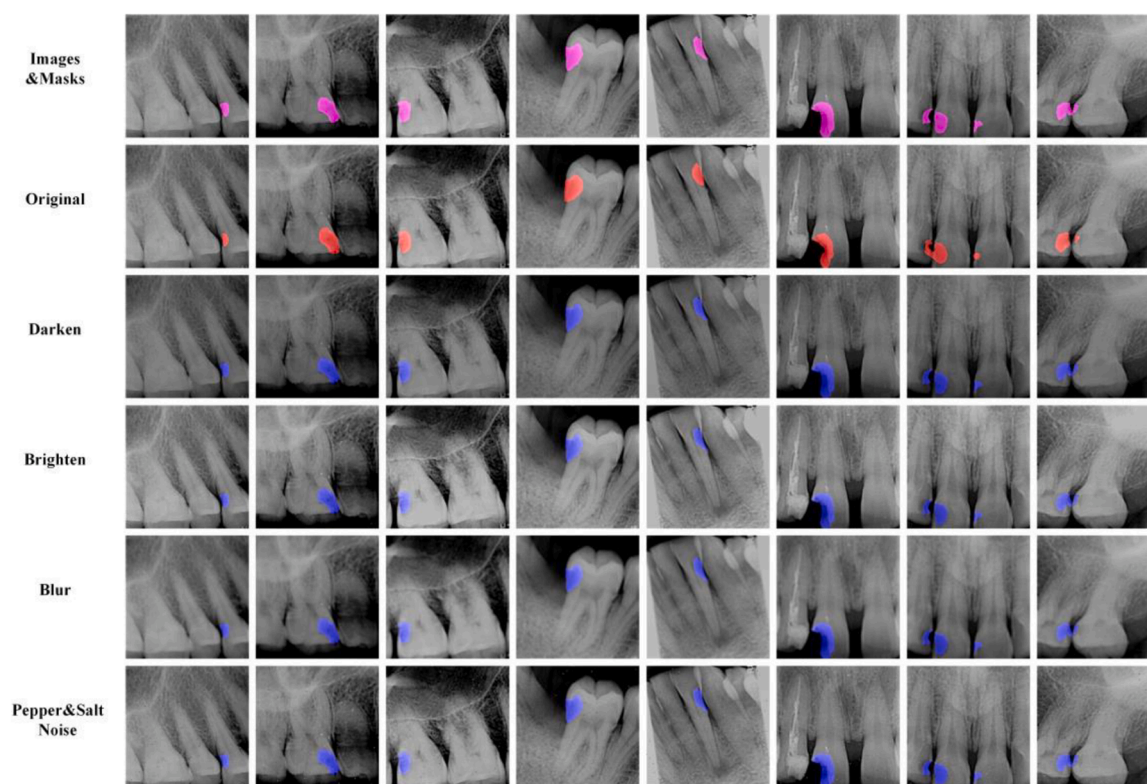


Fig. 7. Example segmentation results in the robustness test of our network.

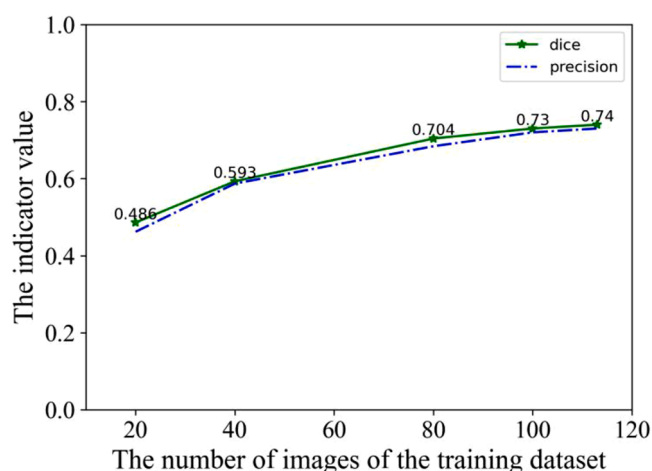


Fig. 8. The performance of our network under different numbers of original training images.

to enhance the multi-scale and global feature extraction capability. By comprehensive comparison with the typical UNet, UNet++, Trans-UNet, and Swin-UNet on the same tooth X-rays image dataset, our network outperforms the other networks in terms of dice similarity and precision metrics. Our study contributes to the promotion of automatic caries segmentation research and has promising application clinically.

Funding

This study was funded by General Scientific Research Project of Education Department of Zhejiang Province (Grant No. Y202147147) and Zhejiang Provincial Natural Science Foundation of China (Grant Nos. LY20E050013 and LZ21F030002).

CRedit authorship contribution statement

Shunv Ying: Conceptualization, Methodology, Resources, Data curation, Writing – review & editing, Funding acquisition. **Benwu Wang:** Software, Writing – original draft, Visualization. **Haihua Zhu:** Resources, Data curation, Validation. **Wei Liu:** Resources, Data curation, Validation. **Feng Huang:** Conceptualization, Methodology, Writing – original draft, Supervision.

Declaration of Competing Interest

The authors declare no competing interests.

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